# numerical integration: trapezoidal and simpson rules

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**ABSTRACT:**

In this note we look at two simple methods to compute definite integrals numericaly, the trapezoidal rule and Simpson's rule. We test them on one example and compare the errors.

## 1. Introduction

Many integrals can not be written with elementary functions. For example the integral of e^(-x^2)
from 0 to 1 has no simple formula, so we need numerical methods. Given a function f on the interval
[a,b] we want to approximate I = integral from a to b of f(x) dx.

## 2) Trapezoidal rule

Divide [a,b] in n equal pieces of width h = (b-a)/n. The trapezoidal rule is :

```
T_n = h * [ f(a)/2 + f(x1) + f(x2) + ... + f(x_(n-1)) + f(b)/2 ]
```

The error is at most (b-a) h^2 / 12 times the maximum of |f'(x)|.

### *Section 3 - Simpson's rule*

If n is even, Simpson's rule is

```
S_n = (h/3) [ f(x0) + 4 f(x1) + 2 f(x2) + 4 f(x3) + ... + 4 f(x_(n-1)) + f(xn) ]
```

and the error is at most (b-a) h^4 / 180 times the maximum of |f''(x)|.

## 4. NUMERICAL EXPERIMENT

We take f(x) = e^(-x^2) on [0,1]. The exact value is 0.7468241328 (to 10 digits).

| n | T_n | S_n | error T | error S |
|---|---|---|---|---|
| 2 | 0.73137 | 0.747180 | 1.5e-02 | 0.00035 |
| 4 | 0.742984 | 0.74685538 | 0.0038 | 3.1e-5 |
| 8 | 0.74587 | 0.746826 | 9.6e-4 | 2.0e-06 |
| 16 | 0.7465846 | 0.74682 | 0.00024 | 1.2E-7 |

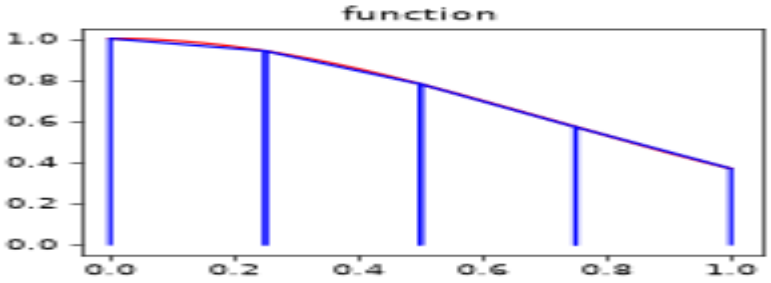



*figure1 - the function and the trapezoids (n=4)*

## 5. conclusion

**Simpson is better than trapezoid. When we double n the error of the trapezoidal rule is divided by 4 and for Simpson by 16 !!!**

**REFERENCES:**

[1] Burden, R. and Faires, J., Numerical Analysis, Cengage

Atkinson K. (1989) An introduction to numerical analysis. Wiley.

2. Press, W.H. et al - Numerical Recipes - 2007 (Cambridge)